Project 2- Rachana Pandey

Project Title- Enhancer Reprogramming in MCF7 Breast Cancer: The Impact of GATA3 Knockdown and JUN Overexpression

Introduction

Breast cancer is one of the most frequent cancers diagnosed in women worldwide. This heterogeneous disease is divided into four principal molecular subtypes by immunohistological technique based on the expression of the estrogen receptor (ER), the progesterone receptor (PR), and the human epidermal growth factor receptor 2 (HER2). These subtypes are luminal A, luminal B, HER2 and triple-negative breast cancer. Estrogen positive represents around 60% of breast cancers. The current treatment for estrogen positive is endocrine therapy. The most common types of endocrine therapy are selective estrogen receptor modulators (SERMs), selective modulators estrogen receptor degraders (SERDs), and aromatase inhibitors. SERMs, such as tamoxifen compete with estrogen by binding to the ER. This binding changes the conformation of the ER ligand-binding domain, and once ER is translocated to the nucleus, it blocks co-factor recruitment and subsequent genes transcription involved in cell cycle progression (cyclin D1), cell proliferation (like IGF-1), or cell migration. It is an adjuvant therapy orally administered for 5 to 10 years according to tumor aggressivity (Burguin et al., 2021). Tamoxifen has been used for the last 40 years as an endocrine therapy for breast cancer. However, resistance against tamoxifen is still one of the major hurdles in the effective management of breast cancer. Resistance leads to the development of other cancers like uterine cancer. Several biochemical factors and molecular pathways have been identified to be involved in the tamoxifen resistance such as alteration of ER signaling, crosstalk between ER and GFR (growth factor receptor) network, downregulation of ER, upregulation of specific GFR, activation of PI3/AKT/mTOR pathway including PTEN inactivation and induction of NFκB signaling (Ali et., al 2016). However, the change in the epigenetic landscape of this cancer is not explored in depth.

GATA binding protein 3 (GATA3) is essential for normal development of the mammary gland and associated with ER-positive breast cancer. Loss of GATA3 has been associated with epithelial-mesenchymal transition (EMT) in experimental studies. Downregulation of GATA3 is associated with poor long-term benefit from tamoxifen in ER-positive breast cancer. Whereas patients with higher expression showed a significant benefit (Sandström et., al 2024). Another transcription factor c-Jun has been found to restore the tamoxifen sensitivity. JUN is AP-1 family transcription factors which regulate various cellular processes, including proliferation and survival. JUN has been found to enhance fatty acid oxidation in tamoxifen-resistant cells, thereby promoting survival and resistance to therapy (Jiang et., al 2023). MCF7 cells are estrogen and progesterone positive breast cancer cells. TamR are breast cancer cell line resistant to tamoxifen. They are derived from MCF7 after long term exposure to tamoxifen leading to resistance. For this project, I am using the ChIP-seq data from four experimental conditions in MCF7cells: Control_shRNA, GATA3_knockdown (GATA3_shRNA), JUN overexpression through doxycycline (JUN_OE_DOX), and both GATA3 knockdown and JUN overexpression with doxycycline (GATA3_shRNA_JUN_OE_DOX). This project investigates the role of

GATA3 knockdown (shRNA) and JUN overexpression in enhancer reprogramming by performing ChIP-seq for H3K27ac (a histone mark associated with active enhancers) in MCF7 cells. H3K27ac is a histone modification associated with active enhancers and transcriptional activation (Trovato et., al 2024). By systematically knocking down GATA3 and overexpressing JUN in MCF7 cells, I aim to investigate their roles in enhancer reprogramming and chromatin accessibility changes. This analysis provides insights into how these transcription factors influence enhancer activity, which may have implications for endocrine resistance mechanisms in breast cancer.

Objectives

General Objectives

To investigate how GATA3 knockdown and JUN overexpression influence enhancer reprogramming by analyzing chromatin accessibility (H3K27ac) changes in MCF7 cells and to determine whether their combined effects are synergistic or compensatory.

Specific Objectives

1. To determine the impact of GATA3 knockdown in chromatin accessibility (H3K27ac) in MCF7 cells.
2. To analyze the impact of JUN overexpression on enhancer remodeling and transcriptional regulation.
3. To evaluate whether combining GATA3 knockdown and JUN overexpression leads to a synergistic effect on enhancer reprogramming in MCF cells.

Hypothesis

My main hypothesis is that the knockdown of GATA3 and overexpression of JUN changes enhancer reprogramming in MCF7 cells. Specifically, the knockdown of GAT3 reduces the chromatin accessibility, JUN overexpression increases the enhancer activity, and the combination of both JUN overexpression and GATA3 knockdown results in synergistic effect in chromatin remodelling.

Research Questions

1. How does GATA3 knockdown (shRNA) affect chromatin accessibility (H3K27ac) in MCF7?
2. How does JUN overexpression influence enhancer activation and chromatin remodeling in MCF7 cells?
3. Does combining GATA3 knockdown + JUN overexpression result in a synergistic enhancer reprogramming effect in MCF7 cells?

Methods and Analysis

For this project, I utilized publicly available ChIP-seq data for H3K27ac from the study- "*Enhancer reprogramming driven by high-order assemblies of transcription factors promotes phenotypic plasticity and breast cancer endocrine resistance*" (GEO accession number

GSE128460). The raw FASTQ files corresponding to four experimental conditions in MCF7 cells (Control_shRNA, GATA3_shRNA, JUN_OE_DOX, and GATA3_shRNA_JUN_OE_DOX) were downloaded for analysis. I performed quality control using FastQC to assess read quality, followed by MultiQC to generate a consolidated report across all samples. Reads were aligned to the human reference genome (hg38) using Chromap. The resulting BED files were converted into SAM format, followed by sorting and indexing using SAMtools. Finally, I converted the processed files into BigWig format for visualization. Peaks were identified using MACS2. FRiP score was calculated to evaluate ChIP enrichment quality across samples. Bedtools was used to identify overlapping and unique peaks between different conditions. And then I used deeptools for pairwise correlation analysis of genome-wide H3K27ac enrichment and heatmap generation.

Results and Discussions

Altogether 61, 354 peaks were detected in total across 4 experimental conditions. These peaks reflect the region of H3k27ac enrichment. The highest number of unique peaks was detected in control_shRNA. Similarly, JUN overexpressed cells have 18699 unique peaks, GATA3_shRNA has 6203 unique peaks, and GATA3_shRNA and JUN_OE has only 6826 unique peaks (Figure 1a.). 61 peaks were shared across all treatment conditions which represent the region of active enhancer regardless of the GATA3 knockdown or JUN overexpression. The combined condition (GATA3_shRNA + JUN_OE) results in an overall lower number of unique peaks, suggesting that combined perturbation does not introduce many new enhancer elements and JUN overexpression may partially compensate for the loss of GATA3. The knockdown of GATA3 reduces the total peaks to 9793 which suggests reduced enhancer activity at a specific locus. Similarly, Jun overexpression also decreased the total peaks compared to control, which might indicate a shift in enhancer activation. However, unique JUN peaks suggest JUN drives activation at new enhancer regions. The heatmap shows H3K27ac signal intensity across different experimental conditions (Figure 1b). Many peaks are lost when GATA3 is knocked down, suggesting that GATA3 loss leads to decreased enhancer activity, aligning with the loss of unique peaks in the Venn diagram. Some enhancers remain unchanged, indicating GATA3-independent regulation at certain loci. JUN overexpression increases enhancer activity, as seen by the stronger red signals. JUN may be activating oncogenic pathways by opening new enhancers or amplifying existing ones. GATA3_shRNA_JUN_OE shows a mixed pattern. Some enhancers are lost due to GATA3 depletion, while others are activated by JUN. This suggests that JUN can partially compensate for GATA3 loss, but not completely.

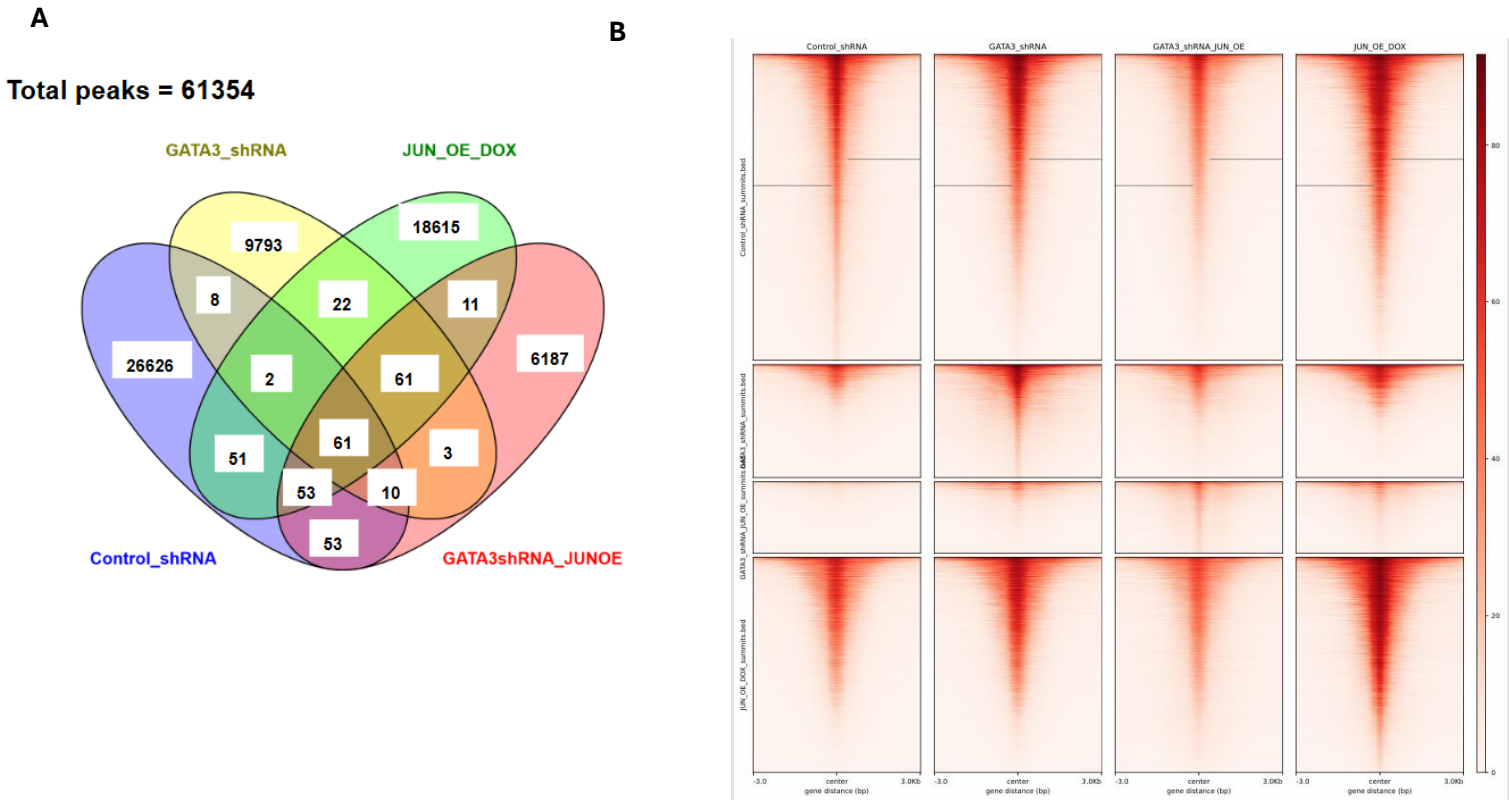


Figure 1- A: Venn Diagram of Shared and Unique H3K27ac Peaks Across Experimental Conditions in MCF7 Cells. B: Heatmap of H3K27ac Signal Intensity Across Experimental Conditions in MCF7 Cells.

The correlation plot quantifies the global similarity between conditions. The scatterplot matrix in figure 2c shows the similarity between the treatment conditions based on H3K27ac signal distribution. The correlation values indicate that Control_shRNA and GATA3_shRNA have a moderate correlation (~0.40–0.43), suggesting that GATA3 knockdown does not drastically alter global H3K27ac levels. JUN_OE_DOX has the lowest correlation with all other conditions (~0.35–0.38), implying that JUN overexpression significantly changes the enhancer landscape which align with the findings from Venn diagram and heatmap in figure 1. The GATA3_shRNA_JUN_OE sample also shows weaker correlations, suggesting that JUN overexpression may override GATA3 effects or that these factors regulate distinct enhancer networks.

Frip score in figure 2a shows the quality metric for the ChIP seq data. It shows the proportion of mapped reads that fall within peaks. A higher FRiP score generally indicates higher ChIP enrichment and better signal quality. Overall, the FRiP score is higher than 0.02 across all

conditions indicating reliable peak detection. Control_shRNA has a relatively high FRiP score, suggesting robust baseline enhancer activity. JUN_OE_DOX has the highest FRiP score, indicating that despite having fewer peaks overall, the H3K27ac signal is highly concentrated within its peaks, suggesting JUN-driven enhancer remodeling or activation at specific loci. GATA3 knockdown has reduced the FRiP scores due to loss of enhancer activity.

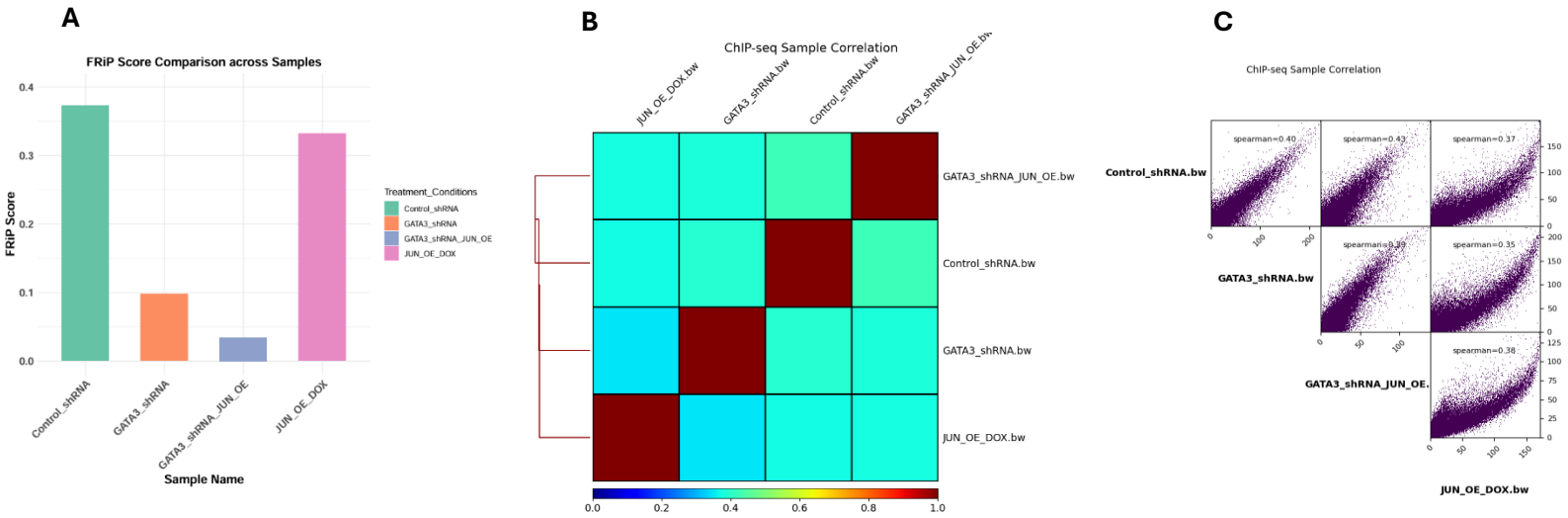


Figure 2- A: FRiP Score Comparison Across Samples. B: Pearson correlation heatmap showing showing H3K27ac enrichment between conditions. C: Spearman correlation scatterplot depicting the relationship between H3K27ac signal intensity across samples.

Conclusion

GATA3 knockdown lead to a loss of H3K27ac suggesting enhancer deactivation. This is in line with our initial hypothesis on GATA3 that GATA3 knockdown decreases chromatin accessibility. However, our hypothesis that JUN overexpression would promote enhancer activation is incorrect. JUN overexpression also decreased enhancer activity. While Jun has lower number of H3k27ac peaks as compared to control, the FriP scores show that the signal is more enriched within those peaks. Also, our initial hypothesis that GATA3 knockdown and JUN overexpression would have a synergistic effect on enhancer activation was not supported. Instead, JUN partially compensated for the loss of GATA3 by restoring H3K27ac at select enhancers, but the two factors do not act in a cooperative manner. Correlation and scatter plot analyses further highlight distinct enhancer landscapes across conditions**,** reinforcing the notion that GATA3 and JUN changes the enhancer programming.

Future Directions

While we found the change in chromatin accessibility due to overexpression of JUN and knockdown of GATA3, we do not know how this affects the gene expression and what will be those affected genes and pathways. Integrating the RNA-seq results to link enhancer activity with gene expression changes to identify functional consequences of JUN and GATA3

perturbations can be a better approach. We can also perform motif enrichment to identify transcription factors binding at JUN and GATA3-dependent enhancers to understand regulatory mechanisms.

References

Ali, S., Rasool, M., Chaoudhry, H., Pushparaj, P. N., Jha, P., Hafiz, A., ... & Jamal, M. S. (2016). Molecular mechanisms and mode of tamoxifen resistance in breast cancer. *Bioinformation*, *12*(3), 135.

Burguin, A., Diorio, C., & Durocher, F. (2021). Breast cancer treatments: updates and new challenges. *Journal of personalized medicine*, *11*(8), 808.

Jiang, C., Zhu, Y., Chen, H., Lin, J., Xie, R., Li, W., ... & Xu, S. (2023). Targeting c-Jun inhibits fatty acid oxidation to overcome tamoxifen resistance in estrogen receptor-positive breast cancer. Cell Death & Disease, 14(10), 653.

Sandström, J., Bomanson, J., Pérez-Tenorio, G., Jönsson, C., Nordenskjöld, B., Fornander, T., ... & Stål, O. (2024). GATA3 and markers of epithelial-mesenchymal transition predict long-term benefit from tamoxifen in ER-positive breast cancer. NPJ Breast Cancer, 10(1), 78.

Trovato, M., Bunina, D., Yildiz, U., Fernandez-Novel Marx, N., Uckelmann, M., Levina, V., ... & Noh, K. M. (2024). Histone H3. 3 lysine 9 and 27 control repressive chromatin at cryptic enhancers and bivalent promoters. *Nature communications*, *15*(1), 7557.